

USER MANUAL

1. Product Info 【必填】

Product Name: Step Down Module

Model: [Insert Model Number]

Batch Number 1005010036500532

2. MANUFACTURER & EU REPRESENTATIVE INFO 【必填】

- Manufacturer: [Guangdong Vilesco Trading Co., Ltd.
- , E2-1204, Building E, Wenchang Garden, West of Wei15 Road, West of Dongcheng District, Yuancheng District, Heyuan City, Guangdong Province, 123204256@qq.com]
- EU Authorized Representative: [MJCM SARL, 78 avenue des Champs-Elysees Bureau 326
- , mjcm190928@gmail.com]

3. Safety & Warnings 【by 类目必填】

Read all instructions before use to ensure safety and prevent damage.

- Warning: Children, Keep the device and its cord out of reach of children under 3 years old.
这块应根据实际商品性质填写，可参考网规[全球速卖通关于售往欧盟的产品警告信息合规须知]
- Read and follow all safety instructions carefully before handling, installing, or using this electronic component. Non-compliance with the following warnings may result in electric shock, fire hazard, product damage, equipment malfunction, or personal injury. All safety guidelines shall be strictly observed throughout product application, maintenance, and storage.

• 1. General Safety Precautions

- This electronic component is designed for professional electronic assembly and industrial application purposes only. It is not intended for consumer DIY unprofessional operation or unauthorized modification. Unauthorized disassembly, cutting, rewelding, or alteration of the component structure and parameters will void all product warranties and may cause unpredictable safety risks. Operators must possess basic electronic operation knowledge and corresponding operational qualifications before handling the product.
- Keep the product away from children and untrained personnel at all times. Do not use the component for any equipment or application beyond its specified electrical

本模版由速卖通提供，仅供参考。卖家应根据产品的实际特性及销往国的法律要求自行调整。速卖通不对因使用本模版产生的任何合规或法律问题承担责任。This template is provided by AliExpress for reference only. Sellers should adjust the content based on actual product features and local laws. AliExpress assumes no responsibility for any compliance or legal issues arising from the use of this template.

parameters and operating scope stated in the product datasheet. Any misuse, overload, or abnormal application is strictly prohibited.

• 2. Electrical Safety Warnings

- Do not apply voltage or current exceeding the component's maximum rated parameters, including rated voltage, rated current, power dissipation, and operating frequency. Overload operation will cause overheating, breakdown, short circuit, component burnout, and even fire or electric shock accidents.
- All installation and welding operations must be performed in a power-off state. Confirm that the circuit power is completely cut off and the capacitor energy is fully discharged before contact with the component pins and circuits. Avoid live operation to prevent electric shock and component electrostatic damage.
- Prevent reverse polarity connection, wrong pin insertion, and short circuit of component pins. Strictly follow the product pin definition, circuit diagram and application guidelines for wiring and assembly. Abnormal circuit connection will lead to permanent component failure and secondary damage to the supporting equipment.

• 3. Electrostatic Discharge (ESD) Protection

- Most electronic components are sensitive to static electricity. Static discharge generated by human body, operating tools and production environment may cause latent or permanent damage to the component, resulting in unstable performance or functional failure.
- All handling, testing and assembly operations must be carried out in a grounded ESD-safe working environment. Operators shall wear qualified anti-static wristbands, anti-static gloves and anti-static clothing. Operating tools, welding equipment and test instruments must be reliably grounded. Unpack the product only after completing all electrostatic protection measures.

• 4. Operating & Environmental Safety

- Operate the component strictly within the specified temperature, humidity, atmospheric pressure and environmental range. Avoid using or placing the product in high temperature, high humidity, dusty, corrosive, flammable and explosive environments, as well as places with strong electromagnetic interference. Extreme environments will accelerate component aging, reduce service life, and induce performance failure and safety accidents.
- Do not expose the component to direct high-temperature baking, sunlight long-term irradiation, water immersion, liquid corrosion and mechanical vibration or impact. Avoid bending, squeezing and scratching the component shell and pins to prevent internal circuit damage and structural failure.

本模版由速卖通提供，仅供参考。卖家应根据产品的实际特性及销往国的法律要求自行调整。速卖通不对因使用本模版产生的任何合规或法律问题承担责任。This template is provided by AliExpress for reference only. Sellers should adjust the content based on actual product features and local laws. AliExpress assumes no responsibility for any compliance or legal issues arising from the use of this template.

- 5. Storage & Transportation Safety

- Store the product in a dry, cool, ventilated and dust-free indoor environment with standard temperature and humidity. Keep away from heat sources, fire sources, chemical corrosives and conductive impurities. Sealed storage is recommended for unused components to prevent moisture absorption, oxidation and pin rust.
- During transportation and handling, avoid severe vibration, collision, extrusion and falling. Do not mix and transport with corrosive, flammable and heavy goods. Maintain complete original packaging protection to ensure component integrity and performance stability.

- 6. Maintenance & Abnormal Handling

- Once the component has abnormal conditions such as overheating, smoke, scorching smell, short circuit failure and performance abnormality during operation, cut off the power supply immediately, stop using it, and isolate the faulty component. Do not continue to use or repair the damaged component without authorization to avoid secondary accidents.
- Product maintenance and inspection must be completed by professional and technical personnel. Discard and dispose of failed and scrapped components in accordance with local electronic waste environmental protection regulations, and do not discard them randomly to avoid environmental pollution and safety hazards.

- 7. Disclaimer

- The company shall not be liable for any product damage, equipment failure, personal injury and economic loss caused by unauthorized modification, improper installation, non-compliant operation, environmental abuse and violation of the above safety warnings. The right of final interpretation of these safety warning terms belongs to the product manufacturer.
-

4. Instructions for Use 【必填】

如：

1. This guideline provides standard operating procedures for the unpacking, installation, assembly, testing and application of this electronic component. To ensure product performance stability, service life and application safety, all operators shall strictly follow the instructions below. These standard instructions apply to professional industrial assembly and engineering matching scenarios of the product.

本模版由速卖通提供，仅供参考。卖家应根据产品的实际特性及销往国的法律要求自行调整。速卖通不对因使用本模版产生的任何合规或法律问题承担责任。This template is provided by AliExpress for reference only. Sellers should adjust the content based on actual product features and local laws. AliExpress assumes no responsibility for any compliance or legal issues arising from the use of this template.

2. 1. Unpacking & Inspection
3. Unpack the product in a clean, dry and ESD-protected working environment. Check the outer packaging integrity first to confirm no damage, extrusion, moisture or contamination during transportation. After unpacking, inspect the component appearance carefully for scratches, pin bending, oxidation, cracking, missing parts or other visual defects.
4. Verify the product model, specification, batch number and quantity are consistent with the purchase requirements and product datasheet. It is prohibited to use any mismatched, damaged or defective components. Please keep the original packaging materials for subsequent storage and return management if needed.

2. Pre-Use Preparation

Confirm all operating tools, welding equipment and test instruments are intact, calibrated and reliably grounded. Operators must complete ESD protection preparation, including wearing anti-static wristbands, anti-static gloves and anti-static garments, to eliminate static electricity from human body and working environment.

Check the working environment to ensure it is free from dust, corrosive gas, condensed water, strong electromagnetic interference and extreme temperature and humidity. Confirm the application circuit design fully complies with the component's rated parameters, application range and circuit matching requirements specified in the official datasheet.

3. Standard Installation & Assembly Steps

All assembly and installation operations must be carried out in a completely power-off state with circuit capacitors fully discharged. Confirm the pin definition, polarity and installation direction strictly according to the product pinout diagram and engineering drawings to avoid reverse installation, wrong insertion and offset mounting.

For welding operation, control the welding temperature and welding time in accordance with the standard parameters of the component. Excessively high temperature or long-time welding is prohibited to prevent thermal damage to the internal chip and packaging structure. Use qualified solder and auxiliary materials compatible with electronic components, and avoid residual solder paste, tin slag and flux contamination on the component surface and pins.

After welding, clean the component and circuit board properly to remove residual flux and impurities, and check for virtual welding, missing welding, short circuit and cold welding. Ensure the component is firmly installed with no loose pins or offset position.

4. Power-On & Testing Operation

Complete a full circuit inspection before power-on, confirm all wiring is correct, no short

本模版由速卖通提供，仅供参考。卖家应根据产品的实际特性及销往国的法律要求自行调整。速卖通不对因使用本模版产生的任何合规或法律问题承担责任。This template is provided by AliExpress for reference only. Sellers should adjust the content based on actual product features and local laws. AliExpress assumes no responsibility for any compliance or legal issues arising from the use of this template.

circuit, open circuit or wrong connection exists, and the component installation position is accurate and firm. Adopt step-by-step power-on mode: firstly apply low voltage for pre-test, check for abnormal heating, current surge and abnormal noise, then adjust to rated working voltage and current after confirming normal status.

Test the product electrical performance, signal stability and functional indicators in accordance with the standard test procedures. All test data shall meet the parameter standards specified in the product datasheet. Stop testing immediately and power off the equipment once any abnormal data or abnormal state is found, and conduct troubleshooting after isolation.

5. Normal Application Requirements

The component shall be used within the rated working voltage, rated current, power consumption, operating temperature and frequency range at all times. Avoid long-term overload operation, frequent sudden start and stop, and abnormal surge impact, so as to ensure stable operation and extend product service life.

Keep the component working environment clean and ventilated during operation to ensure good heat dissipation. Avoid covering, blocking the heat dissipation area, or attaching conductive dust, oil dirt and corrosive substances to the component surface. Regularly check the working state of the component and supporting circuits to eliminate potential risks in advance.

6. Daily Maintenance & Inspection

Carry out regular offline inspection and maintenance for components in long-term operation. Focus on checking pin firmness, surface cleanliness, heat dissipation condition and circuit connection stability. Timely clean surface dust and dirt, and replace aging and failed components in time.

For components in standby state for a long time, regular power-on detection is recommended to avoid performance degradation caused by long-term static placement and moisture absorption. All maintenance operations must be performed in a power-off and fully discharged state by professional personnel.

7. Prohibited Operations

It is strictly prohibited to disassemble, modify, polish or cut the component structure and pins without authorization. Prohibited to use the component beyond the specified application scenarios and parameter limits. Prohibited live installation, live maintenance and irregular welding operation. Prohibited to expose the component to corrosive, flammable, high-pressure and severe vibration environments for a long time.

8. Scrap & Replacement Instructions

Once the component has permanent failure such as breakdown, burnout, structural damage
本模版由速卖通提供，仅供参考。卖家应根据产品的实际特性及销往国的法律要求自行调整。速卖通不对因使用本模版产生的任何合规或法律问题承担责任。This template is provided by AliExpress for reference only. Sellers should adjust the content based on actual product features and local laws. AliExpress assumes no responsibility for any compliance or legal issues arising from the use of this template.

and functional failure, it shall be stopped using and replaced in time. Failed components shall be classified as electronic waste and disposed of in accordance with local environmental protection laws and industry specifications, and shall not be randomly discarded or reused.

- 5. *Preparation: Remove all protective films from the device.*
- 6. *Charging: Connect the provided USB cable to a certified 5V power adapter.*
- 7. *Power On: Press and hold the power button for 3 seconds.*
- 8. *Modes: Short press the button to cycle through functions.*

5. COMPLIANCE INFO 【选填】

如果是电子、玩具、PPE，说明书最后必须印有 CE 标志（高度不小于 5mm）

6. MAINTENANCE & STORAGE 【选填】

如：

- *Cleaning: Turn off the device before cleaning. Use a soft, lint-free cloth.*
- *Storage: Store in a dry place at room temperature (15°C - 35°C).*
- *Long-term Storage: If not used for a long time, charge the device to 50% every 3 months.*

7. DISPOSAL & RECYCLING 【选填】

如：

(WEEE Logo Here)
This symbol indicates that this product should not be treated as household waste. According to the WEEE Directive (2012/19/EU), it must be disposed of at an appropriate electronic waste collection point for recycling to prevent potential negative consequences for the environment and human health.

8. TECHNICAL SPECIFICATIONS 【选填】

Feature	Details
Input Voltage	[e.g., 100-240V ~ 50/60Hz]
Rated Power	[e.g., 10W]

本模版由速卖通提供，仅供参考。卖家应根据产品的实际特性及销往国的法律要求自行调整。速卖通不对因使用本模版产生的任何合规或法律问题承担责任。This template is provided by AliExpress for reference only. Sellers should adjust the content based on actual product features and local laws. AliExpress assumes no responsibility for any compliance or legal issues arising from the use of this template.

Feature	Details
Battery Capacity	[e.g., 2000mAh]
Certifications	CE, RoHS, UKCA

本模版由速卖通提供，仅供参考。卖家应根据产品的实际特性及销往国的法律要求自行调整。速卖通不对因使用本模版产生的任何合规或法律问题承担责任。This template is provided by AliExpress for reference only. Sellers should adjust the content based on actual product features and local laws. AliExpress assumes no responsibility for any compliance or legal issues arising from the use of this template.